

Dr. Akashdeep Akashdeep

Experimental Physicist (PhD) — Thin-Film Process, Characterization & Instrumentation

Mainz, Germany

+49 157 51084397 | ghalayan.akashdeep@gmail.com

LinkedIn: akashdeep-condensed-matter | ORCID: 0009-0001-9970-2437

Profile

PhD experimental physicist who builds and fixes the equipment other people depend on. For about five years, I was the local owner of a €0.5–1.0M thin-film deposition and measurement facility, integrating new instruments, keeping it running, and training 10–12 researchers to operate it independently. I work across **thin-film process development**, **measurement-system building** and **data analysis**, and I am moving that hands-on toolkit from research into industrial R&D and production.

Work Experience

2021–Present

Research Associate

Johannes Gutenberg University Mainz, Germany

Retained as research associate (Wissenschaftlicher Mitarbeiter) after PhD defense, May 2026

- Owned a **€0.5–1.0M** thin-film deposition and measurement facility for about five years as local tool owner — scheduling, maintenance coordination, vendor and service liaison, consumables and safety — and trained **10–12 researchers** into independent users while remaining the escalation point for the difficult problems.
- Developed and optimized **thin-film deposition processes** (magnetron sputtering, PLD), including **200 mm wafer** runs on a production-capable UHV platform, with post-deposition XRR thickness and MOKE uniformity checks and structured eLabFTW process documentation.
- Characterized **400–500 thin-film samples** across structural, magnetic and electrical metrology; used electrical-transport data as a diagnostic to flag process/tool drift and measurement artifacts across deposition runs and device batches.
- Extended and maintained the group's **LabVIEW** measurement-control software — new subVIs, new-instrument integration, redesigned sweep schemes — enabling multi-day unattended cryogenic runs with logging, alarms and remote monitoring.
- Built **multi-TB Python** data workflows (batch processing, metadata schemas, reproducible fitting and plotting, basic ML classification) that collaborators could reuse.
- Coordinated multi-institutional projects across Europe, Asia and the USA, and secured about **€0.2M** of in-kind facility access through 8 days of peer-reviewed beamtime and a DAAD RISE-funded student internship.

2020–2021

Junior Research Fellow

Indian Institute of Technology Delhi, India

- Optimized **Heusler-alloy** (Co_2MnAl) sputtering for repeatable thin-film growth; linked deposition parameters to magnetic anisotropy, transport and reproducibility across sample batches.
- MOKE characterization and sputter-chamber assembly; built a custom cryostat transport insert with wiring, temperature sensing, calibration and LabVIEW-based control.

Education

2021–2026	**PhD, Physics (Experimental Condensed Matter)** Johannes Gutenberg University Mainz, Germany — defended May 2026
2017–2019	**Master of Science, Physics (80.4%)** Indian Institute of Technology Delhi, India
2014–2017	**Bachelor of Science, Physics (75.2%)** University of Delhi, Ramjas College, India

Selected Projects

Three builds where I re-architected the lab rather than just using it. Full portfolio at a-akashdeep.github.io/work/.

RHEED-into-PLD integration. Integrated a reflection high-energy electron diffraction system into a pulsed-laser-deposition chamber end to end: spatial-constraint analysis, multiple CAD layout iterations reviewed with the group head, chamber relocation, a custom support structure and dedicated pumping unit, electron-beam alignment, and commissioning into routine use with operating procedures.

Cryostat sample-stage sensor. The cryostat's thermometry was reading a different temperature than the sample itself. I mounted a calibrated sensor directly at the sample position — wiring, electronics and calibration — which removed a systematic temperature artifact and made the electrical and magnetic data trustworthy.

Vacuum transfer suitcase. Specified and procured the active and passive pumps and a transfer arm, CAD-designed the parts for the workshop, then assembled, tested and calibrated a vacuum-preserving sample-transfer path compatible with both the PLD and sputtering chambers.

Skills & Characterization

Process & deposition

- DC/RF magnetron sputtering
- Pulsed laser deposition (PLD)
- Reactive sputtering, thermal evaporation
- 200 mm UHV wafer processing
- Ion-beam etching (IBE / RIBE)
- Optical / e-beam lithography
- Lift-off, annealing, wire bonding

Characterization & metrology

- XRD / XRR
- AFM, SEM
- SQUID / VSM
- MOKE, FMR
- Cryogenic magnetotransport ($\sim$2 K, high field)
- RF / microwave to $\sim$20 GHz
- Wafer thickness & uniformity mapping

Instrumentation & software

- HV / UHV vacuum systems
- RHEED integration & alignment
- Cryostats & superconducting magnets
- Design-for-build CAD (vacuum hardware)
- Python (pandas, NumPy, matplotlib)
- LabVIEW; scipy.optimize fitting
- HDF5, eLabFTW, OriginPro, KLayout

Achievements

- All-India Rank 35, CSIR-NET 2019 (national physics eligibility / fellowship examination).
- All-India Rank 107, IIT-JAM 2017.
- Top 2% nationally, CBSE Higher Secondary examination.

Languages

English — fluent (C2-level professional) **German** — A2–B1, actively improving **Hindi** — native

References

Univ.-Prof. Dr. Mathias Kläui
JGU Mainz — Research Group Head
klaeui@uni-mainz.de

apl. Prof. Dr. Gerhard Jakob
JGU Mainz — PhD Supervisor
jakob@uni-mainz.de